

STM32Axis Development Board

Custom STM32F401RET6 Development Board made by team

ERROR 505

Project Documentation — Schematic, PCB Design, BOM & Validation

Repository: github.com/tanishka-0/stm32axis

1. Overview

The STM32Axis Development Board is a compact, two-layer STM32F401RET6 evaluation board designed as an independently engineered alternative to the STM32 NUCLEO-F401RE. It integrates the microcontroller, regulated power, external SWD debug access, user controls, and the commonly used communication interfaces (UART, SPI, I²C, GPIO) within a board size of 46 × 45 mm.

This document consolidates the complete design record for the board: problem statement and proposed solution, circuit architecture, functional block diagram, schematic, PCB layout and 3D render, bill of materials, power budget, firmware/usage notes, design considerations, and the electrical/design-rule validation performed on the project.

- **Key Specifications**

Parameter	Value
Microcontroller	STM32F401RET6 (ARM Cortex-M4, 84 MHz, 512 KB Flash, 96 KB SRAM, LQFP- 64)
Board size	46 mm x 45 mm
PCB layers	2 (top signal + bottom GND plane)
Input power	USB VBUS, 5 V
Regulated rail	3.3 V, via LD1117S33CTR (800 mA)
Overcurrent protection	Resettable polyfuse, 24 V max / 500 mA
Clock source	8 MHz external crystal (HSE)
Debug interface	5-pin SWD header (external ST-LINK)
Peripheral interfaces	UART, SPI, I2C, GPIO expansion (PC2-PC9)
User I/O	1 user button, 1 reset button, power LED, user LED
Track width / clearance	0.20 mm / 0.20 mm
Via size / drill	0.60 mm / 0.30 mm

- **Problem Statement**

The objective is to design a compact and functional STM32-based evaluation and development board that provides the essential facilities required for programming, debugging, peripheral interfacing and general-purpose embedded development. The design is inspired by the functionality and general architecture of the STM32 NUCLEO-F401RE while requiring an independently created schematic and PCB layout. The board must integrate the MCU, regulated power, programming/debug access, user controls and commonly used communication interfaces within a compact two-layer PCB.

- **Proposed Solution**

A custom development board centered on the STM32F401RET6 ARM Cortex-M4 microcontroller is proposed. USB Micro-B provides the primary 5 V input and USB data connection. The input passes through a resettable polyfuse and is converted to a regulated 3.3 V rail using an LD1117S33CTR linear regulator. The MCU is

programmed and debugged through an external ST-LINK using a dedicated SWD header, avoiding the additional circuitry and board area required for an onboard debugger.

The board also exposes UART, SPI and I²C interfaces and provides GPIO expansion, a user push button, reset control, power and user LEDs and an external 8 MHz crystal — creating a compact platform suitable for firmware development, peripheral experiments and embedded-system prototyping.

• Circuit Architecture

1. PowerSupply Chain

USB VBUS (5 V) -- resettable polyfuse -- 3.3 V LDO (LD1117S33CTR) -- common 3.3 V rail.

Regulator input/output capacitors support stable operation, while dedicated decoupling capacitors are provided at each MCU supply pin.

2. MCU and Clock

The STM32F401RET6 is the central processing device. An 8 MHz external crystal connected to the HSE oscillator pins provides the external high-speed clock source, with dedicated load capacitors placed close to the oscillator pins.

3. Programming and Debugging

PA13/SWDIO and PA14/SWCLK are routed to a 5-pin SWD header together with target 3.3 V, NRST and GND, for connection to an external ST-LINK debugger.

4. USB and Peripheral Interfaces

- USB: D- and D+ connected to PA11 and PA12 respectively.
- UART: PA2 / PA3.
- SPI: PA5 / PA6 / PA7 with a chip-select connection.
- I²C: PB6 / PB7, with external 4.7 kΩ pull-ups.
- GPIO expansion: PC2–PC9 exposed alongside 3.3 V and GND.

5. Functional Block Diagram

The diagram below summarises the board's functional architecture: the USB power/data entry point, the fuse and linear-regulator power chain feeding the 3.3 V rail, the STM32F401RE at the centre, the 8 MHz clock source, user-facing controls and indicators (user button, reset, power LED, user LED), and the peripheral headers (UART, SPI, I²C, SWD, GPIO expansion) fed from the MCU and the 3.3 V rail.

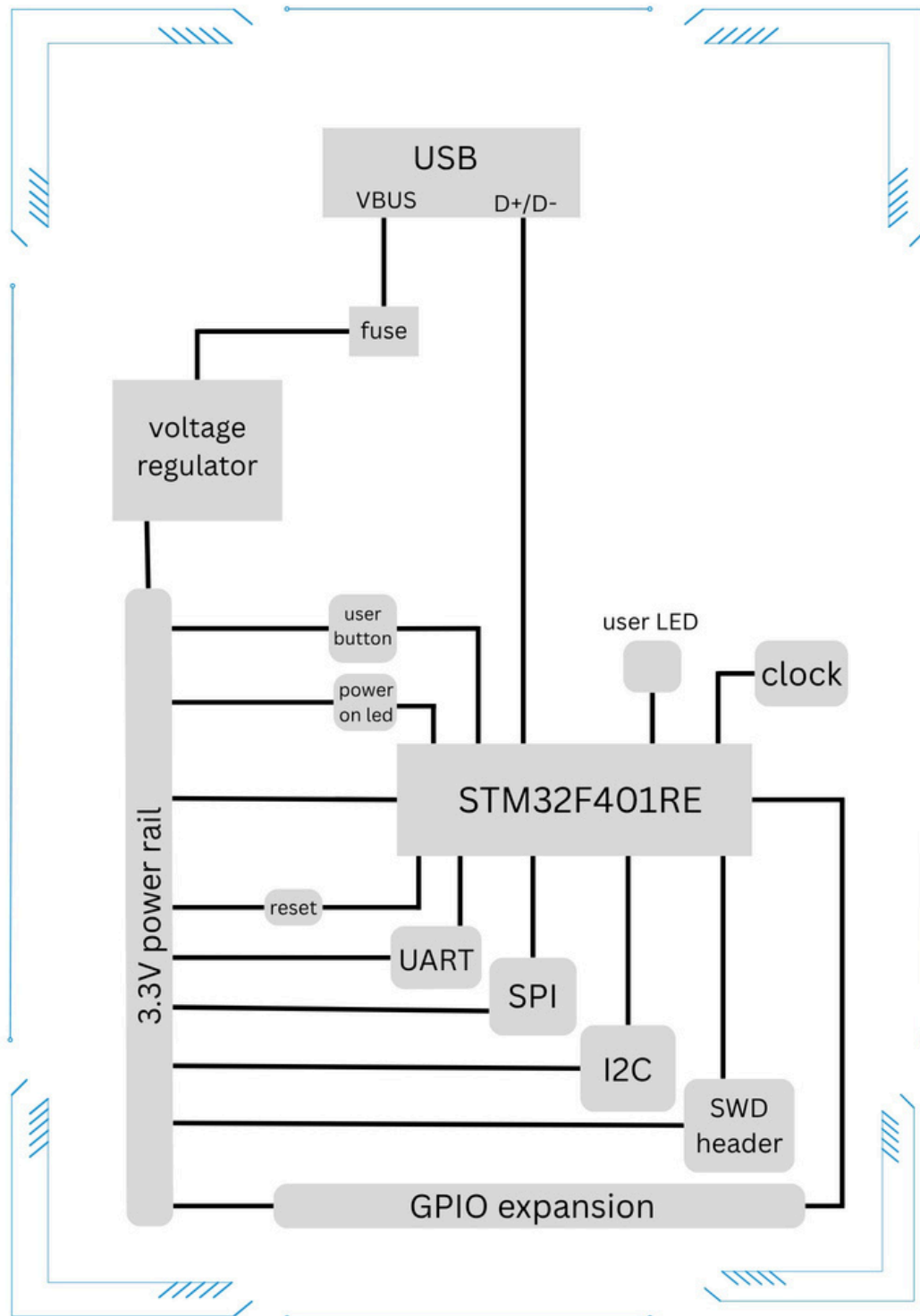


Figure — Functional block diagram of the STM32Axis Development Board

6. Schematic

The complete circuit schematic shows the STM32F401RET6, power regulation stage, USB interface, clock circuit, SWD debug header, and all peripheral connectors (UART, SPI, I²C, GPIO). The full schematic (KiCad) is included in the project repository.

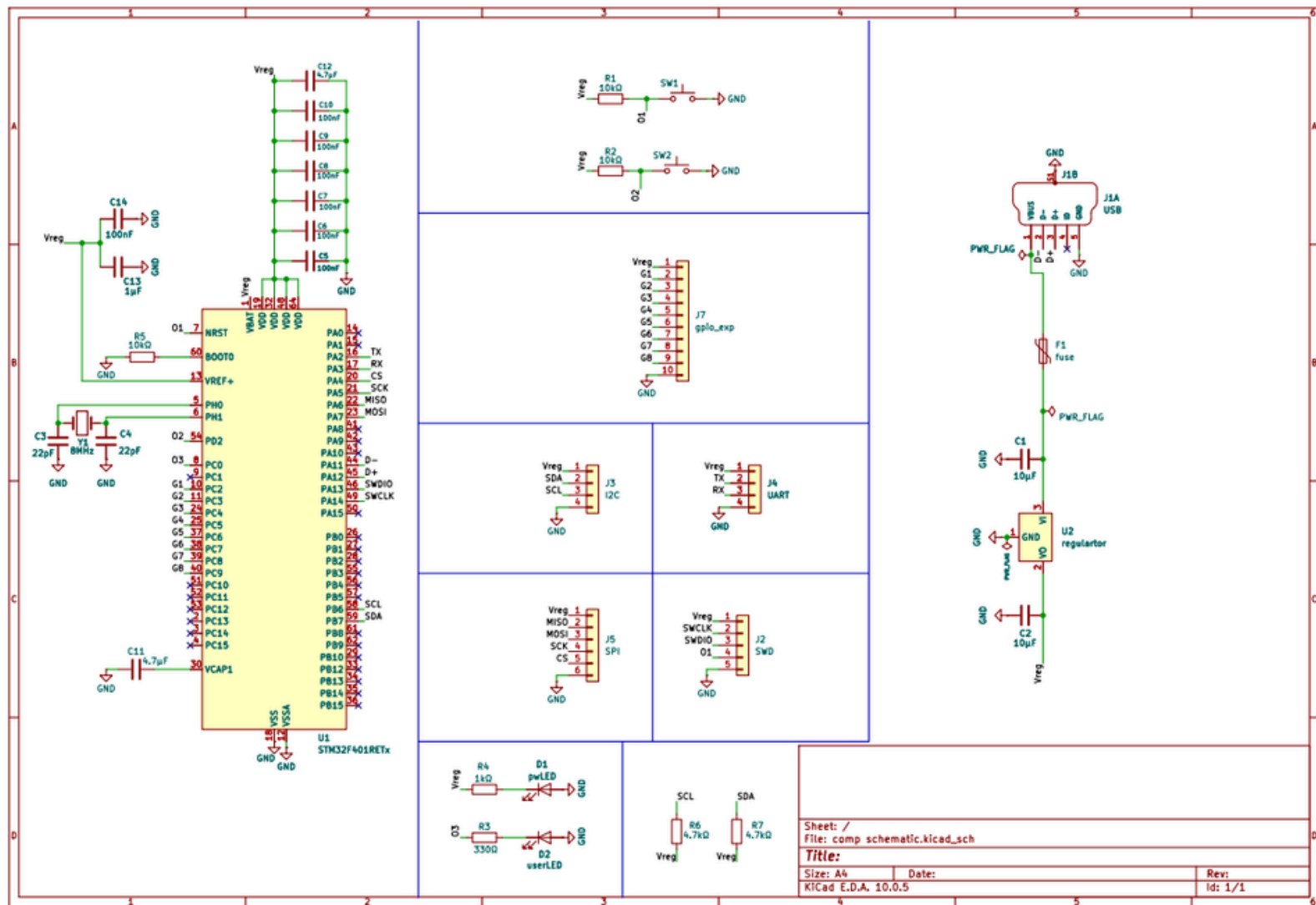


Figure - Schematic of the STM32Axis Development Board

• PCB Layout

The PCB is implemented as a two-layer board with the STM32F401RET6 placed near the centre and interface headers arranged around the board perimeter for accessibility. The USB connector and power-regulation section are placed together to keep the power path compact. The SWD header is positioned for connection to an external programmer/debugger, while UART, SPI, I²C and GPIO headers provide organised peripheral access.

The bottom copper layer is used substantially as a GND plane to provide a low-impedance return path and support signal integrity. Local decoupling components are placed close to the MCU supply pins. Routing was performed with a 0.20 mm track width and 0.20 mm clearance, with compact 0.60 mm vias (0.30 mm drill) where required. Silkscreen labels identify the major interfaces, controls and GPIO pins to improve usability; a '●' marks pin 1 on each communication connector for orientation.

3D Render

A 3D render of the assembled board is included in the project repository, showing component placement for the MCU, USBconnector,crystal, buttons, LEDs and header rows.

• Bill of Materials

The board is populated using standard 0805 passives and readily available connectors, keeping the design low-cost and easy to assemble. The full component list is summarised below; the complete BOM with vendor and pricing detail is included in the project repository.

Sl. No	Reference Designator	Component Description	Value/Rating	Package Type	Brand/Manufacturer	Vendor/Source	Part Number	Total Qty	Rate(Rs)
1	D1, D2	0805 Surface Mount LED Red	30 mA max	0805, SMD	Generic	ROBU	nil	2	2.88
2	SW1, SW2	Tactile Switch SPST-NO Top Actuated Surface Mount	50 mA, 12V	4.5x4.5 mm SMD	C&K	DigiKey	PTS647SN38SMTR2 LFS	2	57.34
3	U1	ARM Cortex-M4 32-bit Microcontroller	84MHz 512 KB Flash 96 KB SRAM, 64pins	LQFP-64	STMicroelectronics	Evelta	STM32F401RET6	2	342.2
4	J1	Micro USB Connector B Female	3.3V	5 Pin SMD, mid mount 4 mounting THT	Molex	Sunrom Electronics	47642-0001	2	27.6
5	J2	Female Straight Header Strip	1×5, 2.54 mm	Through hole(vertical)	Sunrom	Sunrom Electronics	7846	2	5.29
6	J3, J4	Female Straight Header Strip	1×4, 2.54 mm	Through hole(vertical)	Sunrom	Sunrom Electronics	7845	2	4.72
7	J5	Female Straight Header Strip	1×6, 2.54 mm	Through hole(vertical)	Sunrom	Sunrom Electronics	7847	1	5.75
8	J7	Female Straight Header Strip	1×10, 2.54 mm	Through hole(vertical)	Sunrom	Sunrom Electronics	7850	1	7.59
9	U2	Fixed 3.3 V Low-Dropout Linear Voltage Regulator (LDO)	3.3V, 800mA	SOT-223-4	STMicroelectronics	ROBU	LD1117S33CTR	1	29
10	F1	PPTC Resettable Fuse, 1206	24V max, 500mA	1206 SMD	WEITE	Sharvi Electronics	WT1206-050	1	12.35
11	Y1	8MHz Surface Mount Crystal	8MHz, 20pF	HC-49S-SMD	YXC Crystal Oscillators	ROBU	X49SM8MSD2SC	1	13
12	R1, R2, R5	10K Ohm 1/8 Watt Resistor	10K Ohm ±1% tolerance	0805, SMD	YAGEO	Sharvi Electronics	RC0805FR-0710KL	3	2.46
13	R3	330 Ohm 1/8 Watt Resistor	330 Ohms Tolerance: ±1%	0805, SMD	YAGEO	Sharvi Electronics	RC0805FR-07330RL	1	0.86
14	R4	1K Ohm 1/8 Watt Resistor	1K Ohm Tolerance: ±1%	0805, SMD	Royal Ohm	Sharvi Electronics	0805S8F1001T5E	1	0.82
15	R6, R7	4.7K Ohm, 1/8 Watt Resistor	4.7K Ohm, ±1% tolerance	0805, SMD	Watts	Sharvi Electronics	CR-05FTP-4K7	2	1.64
16	C1, C2	10 µF Multilayer Ceramic Capacitor (MLCC)	10 µF, 16 V, ±10%	0805, SMD	YAGEO	Sharvi Electronics	CC0805KKX7R88B106	2	5.44
17	C5, C6, C7, C8, C9, C10, C14	100nF Multilayer Ceramic Capacitor (MLCC)	100nF, 50 V, ±10%	0805, SMD	YAGEO	Sharvi Electronics	CC0805KKX7R98B104	7	13.79
18	C11, C12	4.7 µF Multilayer Ceramic Capacitor (MLCC)	4.7 µF, 25 V, ±10%	0805, SMD	Watsin	Sharvi Electronics	0805X475K250CT	2	4.24
19	C13	1µF Multilayer Ceramic Capacitor (MLCC)	1 µF, 16 V, ±10%	0805, SMD	YAGEO	Sharvi Electronics	CC0805KKX7R78B105	1	2.15
20	C3, C4	22 pF Ceramic Capacitor, C0G/NP0	22 pF, 100 V, ±10%	0805, SMD	YAGEO	Sharvi Electronics	CC0805CRNP00B221	2	4.5
TOTAL									543.62

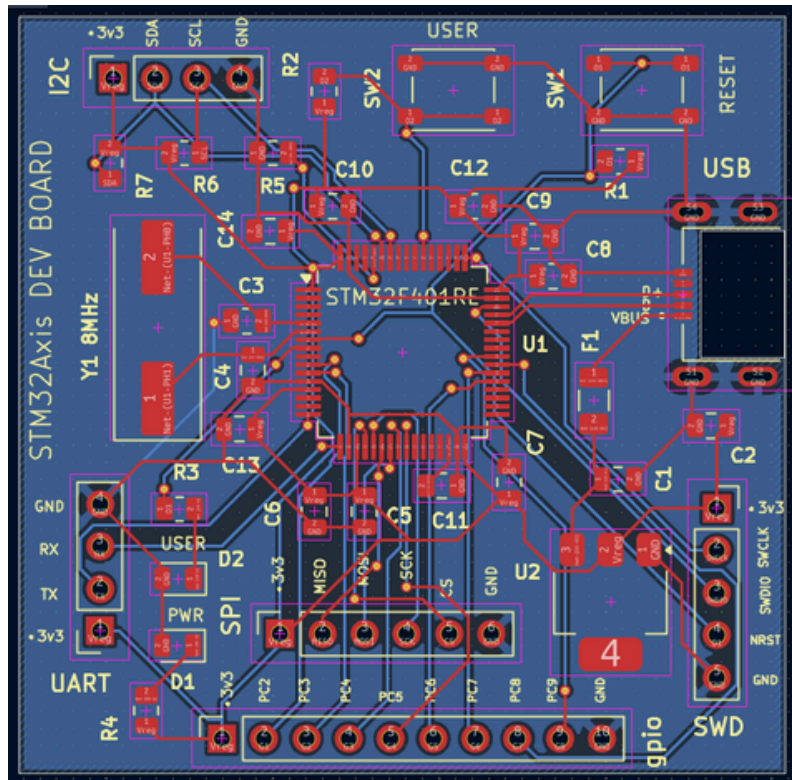


Figure — PCB layout of the STM32Axis Development Board

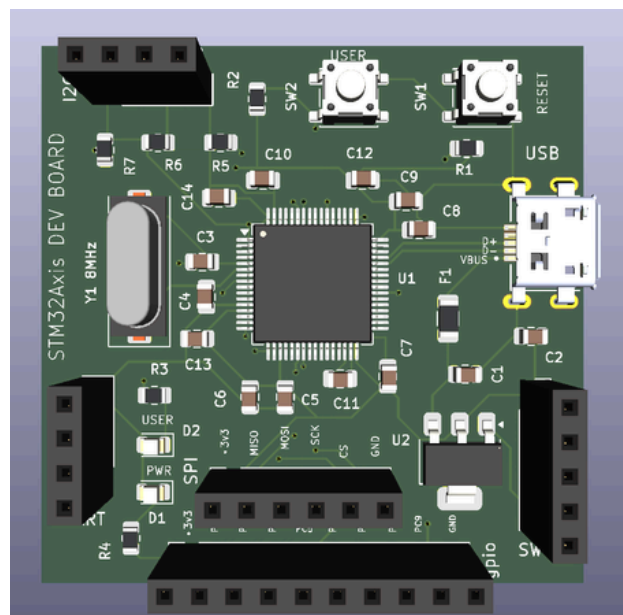


Figure — 3D render image of the STM32Axis Development Board

• Major Components Summary

Component	Function
STM32F401RET6	Main 32-bit ARM Cortex-M4 MCU, 84 MHz, 512 KB Flash, 96 KB SRAM, 64 pins
LD1117S33CTR	3.3 V, 800 mA linear regulator
Micro-USB B connector	USB power input and USB D+/D- interface
Resettable polyfuse	USB input overcurrent protection
8 MHz crystal	External clock source
C&K PTS647SN38SMTR2-LFS	Reset and user push buttons
SWD / UART / SPI / I2C / GPIO headers	Programming, debugging and peripheral expansion
Power & user LEDs	Power indication and GPIO-controlled visual output
Decoupling capacitors	Supply filtering, VCAP support and crystal loading

• Power Budget / Current Estimate

The board has not yet been bench-measured on a physical prototype (see Section 12, Testing/Validation), so the figures below are calculated estimates based on component datasheets and the schematic values, intended to size the regulator margin rather than to represent measured results.

Estimated Load Current

Load	Basis	Typical current
STM32F401RET6 (run mode, 84 MHz, HSE)	Datasheet run-mode supply current at 84 MHz with peripherals active	~30 - 50 mA
Power LED (D1) + R3 (330 Ω)	$(3.3\text{ V} - \sim 2.0\text{ V Vf}) / 330\ \Omega$, continuously on while powered	~4 mA
User LED (D2) + R4 (1 k Ω)	$(3.3\text{ V} - \sim 2.0\text{ V Vf}) / 1\text{ k}\Omega$, only while GPIO drives it high	~1.3 mA
I ² C pull-ups (R6, R7, 4.7 k Ω)	Sinks current only while the bus line is held low	< 1 mA (bus-dependent)
User / reset buttons, crystal load network	Static pull resistors and oscillator loop, negligible steady draw	< 0.5 mA
GPIO expansion / peripheral headers	Depends entirely on what the user connects	Application-dependent

• Design Considerations

- **Power:** The 3.3V rail is regulated from USB input and supported by local bulk and high-frequency decoupling.
- **MCU supply requirements:** The STM32F401RET6 supply, VCAP1 and analog/reference decoupling follow the device hardware recommendations.
- **Clock integrity:** The 8 MHz crystal and its load capacitors are kept close to the oscillator pins and routed with short connections.
- **Debug accessibility:** SWD signals and target-voltage reference are exposed on a dedicated header for external ST-LINK use.
- **Signal integrity & manufacturability:** Controlled routing rules, a continuous GND plane, appropriate clearances and compact via dimensions were used.
- **Usability:** Interface headers and silkscreen labels are arranged to make common development connections easy to identify, with a '1' marking pin 1 on each connector.
- **Compactness:** Functional blocks are placed to minimise unnecessary routing while maintaining accessibility and electrical separation.

• Testing / Validation

Validation was carried out at the design level using schematic connectivity checks (ERC) and PCB design-rule verification (DRC). Functional connections for USB, SWD, reset, clock, power and peripheral headers were also reviewed against the intended circuit architecture and STM32F401RE device requirements. As the project submission is based on PCB design, validation here refers to schematic and layout verification rather than physical prototype measurements.

Electrical Rule Check (ERC)

Metric	Result
Errors	0
Warnings	0
Sheet(s) checked	1 (root sheet)

The ERC run found zero errors and zero warnings, confirming that all electrical connections in the schematic are complete and correctly defined.

Design Rule Check (DRC)

Metric	Result
Errors	0
Violations	1 (footprint mismatch, non-electrical)
Unconnected pads	0
Footprint errors	0

The DRC run reported a single, non-electrical `lib_footprint_mismatch` warning for the USB connector footprint (local footprint override vs. library copy) — a labelling difference only, with no impact on connectivity. Zero unconnected pads and zero footprint errors were found; full DRC and ERC reports are provided in the project repository.

• Peripheral Pin Mapping

The table below summarises the MCU pins routed to each connector, taken from the schematic. Exact net names for the user button and user LED GPIOs should be confirmed against the schematic/netlist in the repository before writing firmware, as this summary does not repeat every housekeeping net.

Interface	Header	MCU Pin(s)	Notes
Debug (SWD)	J2	PA13 (SWDIO), PA14 (SWCLK), NRST	External ST-LINK required
UART	J4	PA2 (TX), PA3 (RX)	3.3 V logic levels
SPI	J5	PA5 (SCK), PA6 (MISO), PA7 (MOSI) + CS	Single chip-select routed
I ² C	J3	PB6 (SCL), PB7 (SDA)	External 4.7 kΩ pull-ups on-board
USB	J1	PA11 (D-), PA12 (D+)	Used for 5 V input; no USB peripheral firmware implemented on this design
GPIO expansion	J7	PC2 - PC9, plus 3.3 V and GND	General purpose; free for application use
Reset	SW1	NRST	Active-low hardware reset

• Challenges

The main design challenges were fitting the required functionality into a compact two-layer board, maintaining adequate clearance around the LQFP-64 MCU, routing multiple peripheral interfaces and providing a practical ground-plane arrangement. Additional care was required for the MCU power and VCAP connections, USB routing, oscillator placement and external SWD accessibility. PCB rule violations encountered during layout were corrected by adjusting routing and clearances while retaining the intended functional architecture.

• Future Scope / Improvements

- Onboard debugger: integrate an ST-LINK-compatible programmer on the board itself (as on the NUCLEO series) to remove the dependency on an external SWD probe.
- USB protection: add ESD protection diodes and common-mode choke on the D+/D- lines, and reverse-polarity/overvoltage protection on VBUS.
- Power options: add a LiPo battery input with charge management for portable/battery-powered use cases, and a power-path switch between USB and battery.
- Expanded peripheral breakout: label and expose additional timer/PWM-capable and ADC-capable pins, and consider a CAN transceiver footprint or SD-card slot for more advanced projects.
- Stack-up: consider a 4-layer PCB with dedicated power and ground planes if higher-speed peripherals (e.g. USB high-speed, higher-rate SPI) are added, for improved EMI and signal integrity.

• Final Outcome

The resulting design is a compact STM32F401RET6 development board, sized 46 × 45 mm, that combines regulated 3.3 V power, USB connectivity, external SWD programming/debugging, clock circuitry, reset and user controls, status indication, and UART, SPI, I²C and GPIO expansion in a two-layer PCB. The design is organised for practical embedded development while retaining a compact form and clear user-facing interface labelling.

The completed schematic, PCB layout, BOM, verification reports and visual renders are available in the project repository:

<https://github.com/tanishka-0/stm32axis>